
FINAL MACHINE LEARNING PROJECT REPORT

Customer Retail Data Analysis

Using Machine Learning Models

Project by

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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions. In this project, different Machine Learning algorithms are implemented using a customer retail dataset. The project focuses on preprocessing data, training models, evaluating performance, and comparing algorithms.

2. Project Objective

The main objective of this project is to implement and compare multiple Machine Learning classification algorithms using a customer retail dataset.

The objectives include:

- ☒ Data preprocessing
- ☒ Handling missing values
- ☒ Encoding categorical data
- ☒ Data visualization
- ☒ Model training and prediction
- ☒ Model evaluation
- ☒ Performance comparison

3. Dataset Description

The dataset used in this project is a customer retail dataset containing sales transaction information.

Features Used

- ☒ Quantity
- ☒ UnitPrice
- ☒ Country

Other Dataset Columns

- ☒ InvoiceNo
- ☒ StockCode
- ☒ Description
- ☒ InvoiceDate
- ☒ CustomerID

```
[6] #CSV File Upload
from google.colab import files
uploaded = files.upload()

... Choose Files customer_r...v file (1).csv
customer_retail csv file (1).csv(text/csv) - 46133245 bytes, last modified: 5/31/2026 - 100% done
Saving customer_retail csv file (1).csv to customer_retail csv file (1) (1).csv
```

```
[8] #Load Dataset
df = pd.read_csv("customer_retail csv file (1).csv")
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850.0	United Kingdom

4. Tools and Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

```
[7] #Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, confusion_matrix
```

5. Data Preprocessing

Data preprocessing is an important step in Machine Learning. The following preprocessing techniques were applied:

5.1 Handling Missing Values

Missing values were removed using the `dropna()` function.

```
[10] #Handle Missing Values
df.isnull().sum()

#Remove missing values
df = df.dropna()
```

5.2 Encoding Categorical Data

The Country column contained text values. LabelEncoder was used to convert text into numerical values.

```
[11]: #Encode Country Column
encoder = LabelEncoder()

df['Country'] = encoder.fit_transform(df['Country'])

df.head()
```

	Quantity	UnitPrice	Country
0	6	2.55	36
1	6	3.39	36
2	8	2.75	36
3	6	3.39	36
4	6	3.39	36

5.3 Feature Selection

The following features were selected:

- Quantity
- UnitPrice
- Country

5.4 Target Creation

A target column was created:

- Quantity > 10 → 1
- Quantity <= 10 → 0

```
[12]: #Create Target Column
df['Target'] = np.where(df['Quantity'] > 10, 1, 0)

df.head()
```

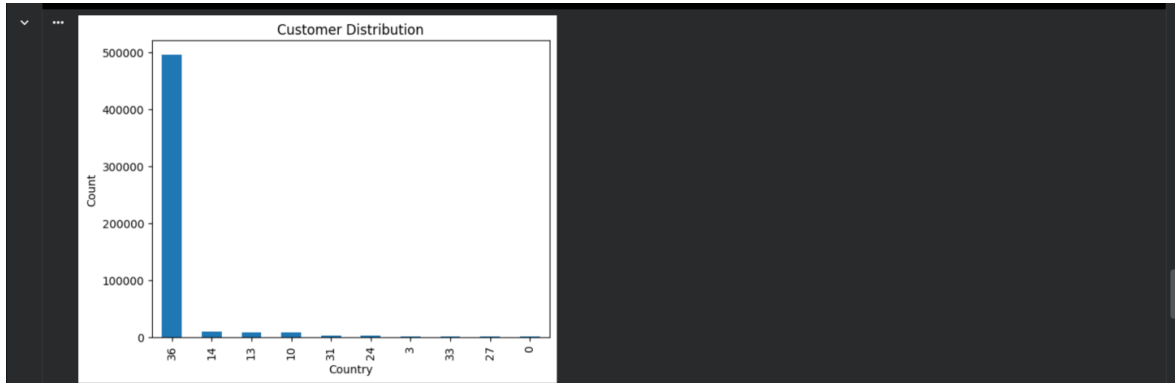
	Quantity	UnitPrice	Country	Target
0	6	2.55	36	0
1	6	3.39	36	0
2	8	2.75	36	0
3	6	3.39	36	0
4	6	3.39	36	0

6. Data Visualization

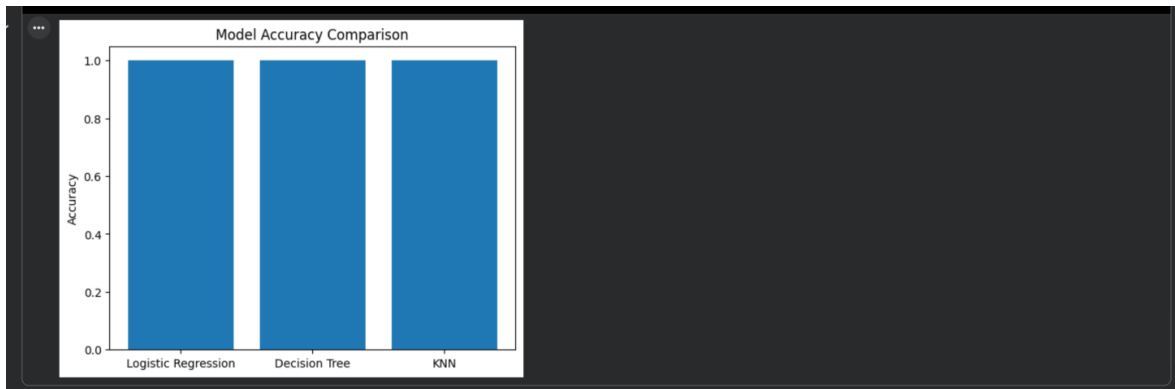
Visualization helps understand the dataset distribution. The graphs were generated using Matplotlib.

Visualizations Included:

📊 Customer Distribution Graph



📊 Model Accuracy Comparison Graph



7. Machine Learning Models Used

7.1 Logistic Regression

Logistic Regression is a supervised learning classification algorithm used for binary classification problems.

7.2 Decision Tree Classifier

Decision Tree is a tree-based Machine Learning algorithm used for classification and prediction.

7.3 K-Nearest Neighbors (KNN)

KNN classifies data points based on the nearest neighboring data samples.

8. Model Training and Testing

The dataset was divided into:

- ✎ 80% Training Data
- ✎ 20% Testing Data

The `train_test_split()` function was used for splitting the dataset.

9. Evaluation Metrics

The following evaluation metrics were used:

Accuracy Score

Accuracy measures how correctly the model predicts the output.

Confusion Matrix

Confusion Matrix shows the classification performance of the model.

10. Results

After training all models, the accuracy scores were compared.

Model	Accuracy
Logistic Regression	1.0
Decision Tree	1.0
KNN	0.9999907733756528

Logistic Regression Accuracy

```
[15] ✓ 2s
#Logistic Regression
log_model = LogisticRegression()

log_model.fit(X_train, y_train)

log_pred = log_model.predict(X_test)

log_acc = accuracy_score(y_test, log_pred)

print("Logistic Regression Accuracy:", log_acc)

print(confusion_matrix(y_test, log_pred))

... Logistic Regression Accuracy: 1.0
[[82035  0]
 [  0 26347]]
```

Decision Tree Accuracy

```
[16] ✓ 0s
#Decision Tree
dt_model = DecisionTreeClassifier()

dt_model.fit(X_train, y_train)

dt_pred = dt_model.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)

print("Decision Tree Accuracy:", dt_acc)

print(confusion_matrix(y_test, dt_pred))

... Decision Tree Accuracy: 1.0
[[82035  0]
 [  0 26347]]
```

KNN Accuracy

```
[17] ✓ 15s
#KNN Model
knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)

knn_pred = knn_model.predict(X_test)

knn_acc = accuracy_score(y_test, knn_pred)

print("KNN Accuracy:", knn_acc)

print(confusion_matrix(y_test, knn_pred))

... KNN Accuracy: 0.9999907733756528
[[82035  0]
 [  1 26346]]
```

11. Conclusion

This project successfully implemented multiple Machine Learning models using a customer retail dataset. Data preprocessing, visualization, model training, prediction, and evaluation were completed successfully.

The project helped in understanding:

- ☞ Supervised learning
- ☞ Classification algorithms
- ☞ Data preprocessing
- ☞ Model evaluation
- ☞ Performance comparison

Different algorithms produced different accuracy results, demonstrating the importance of selecting suitable models for Machine Learning tasks.

12. References

- ☞ Scikit-learn Documentation
- ☞ Google Colab
- ☞ Python Documentation
- ☞ Pandas Documentation